

# Shear-Proof Design Space

Scaling Stirred-Tank Bioreactors for Cell Culture Processes

Engineering Guidance for a Power-Input-per-Unit-Volume (P/V) Calculation Tool

**Source article:** Muralidharan, N. "Shear-Proof Design Space: Scaling Stirred-Tank Bioreactors for Cell Culture Processes." *BioProcess International*, 9 February 2023.

**Compiled:** 27 June 2026 | Purpose: reference basis for a web-based P/V calculation tool. Equation forms below are reconstructed from standard stirred-tank engineering relationships; the source presents several equations as figures, so verify numerical coefficients against the original before production use.

## 1. Purpose & Scaling Philosophy

Scaling a stirred-tank bioreactor (STBR) for mammalian cell culture means holding **scale-independent** parameters constant — pH, temperature, and dissolved oxygen (DO) — while adjusting the **nonlinear, scale-dependent** variables, principally impeller agitation and gas-flow rate. The central engineering challenge is to deliver enough mixing and oxygen transfer to satisfy cell demand while keeping hydrodynamic and bubble-burst **shear** below cell-damaging thresholds.

Kinetic energy supplied by the rotating impeller and by sparged gas is ultimately dissipated as heat. The local **energy dissipation rate (EDR)** drives shear, so a robust “shear-proof” design space is defined by a set of normalized engineering parameters that each remain inside a safe window across scales and geometries.

### Key takeaway

- **P/V** (power per unit volume) and **VVM** (gas volume per liquid volume per minute) are the two most widely used scale-up criteria — but neither captures *localized* shear heterogeneity inside the vessel.
- A complete design space must additionally constrain impeller-zone EDR, tip speed, Kolmogorov microeddy length, gas-entrance velocity, and orifice Reynolds number.

## 2. Impeller-Side Parameters

### 2.1 Power per Unit Volume (P/V)

The primary scale-up criterion. Impeller power draw is normalized by liquid volume so that agitation intensity is comparable across vessel sizes.

$$P = N_p \cdot \rho \cdot N^3 \cdot D^5 \Rightarrow P/V = (N_p \cdot \rho \cdot N^3 \cdot D^5) / V$$

| Symbol | Meaning                                                        | Units                    |
|--------|----------------------------------------------------------------|--------------------------|
| P      | Impeller power input                                           | W                        |
| $N_p$  | Power number (dimensionless; specific to each impeller design) | –                        |
| $\rho$ | Liquid density                                                 | kg/m <sup>3</sup>        |
| N      | Impeller rotational speed                                      | rev/s (s <sup>-1</sup> ) |
| D      | Impeller diameter                                              | m                        |

| Symbol | Meaning               | Units          |
|--------|-----------------------|----------------|
| V      | Liquid working volume | m <sup>3</sup> |

**Constraint:** keep average P/V at or below ~50 W/m<sup>3</sup> to avoid deleterious effects on the cells. **Validity condition:** the impeller Reynolds number must exceed 10,000 (fully turbulent) for N<sub>p</sub> to be independent of speed — a prerequisite for using P/V as a scale-up strategy.

## 2.2 Impeller Reynolds Number (N<sub>Re</sub>)

Ratio of inertial (impeller drag) to viscous forces; confirms the turbulent regime in which P/V scaling is valid.

$$N_{Re} = (\rho \cdot N \cdot D^2) / \mu$$

μ = dynamic viscosity (Pa·s). **Constraint:** N<sub>Re</sub> > 10,000 ⇒ fully turbulent regime.

## 2.3 Impeller-Zone Energy Dissipation Rate (ε<sub>Imp,Zone</sub>)

Average P/V hides the fact that power is not dissipated uniformly. For a pitched-blade impeller, roughly 70.5% of total power is dissipated in the liquid volume immediately swept by the impeller. The localized (impeller-zone) EDR is therefore the more conservative shear metric.

$$\epsilon_{Imp,Zone} = (\text{fraction} \cdot P) / (\rho \cdot V_{Imp,Zone}) \text{ [m}^2/\text{s}^3\text{]}$$

**Constraint:** studies report no lethal effects up to a maximum local energy input as high as 1 × 10<sup>5</sup> W/m<sup>3</sup> (when shear-protecting surfactant is present). Treat this as an extreme upper bound, not a routine setpoint.

## 2.4 Impeller Tip Speed (v<sub>tip</sub>)

The highest fluid velocity occurs at the blade tip; tip speed is a simple, geometry-aware shear surrogate.

$$v_{tip} = \pi \cdot N \cdot D \text{ [m/s]}$$

**Constraint:** v<sub>tip</sub> < 1.5 m/s to prevent adverse effects on cell growth.

## 2.5 Shear Rate at the Blade Tip

Estimated using a drag-coefficient (C<sub>d</sub>) formulation that depends on impeller flow type.

| Impeller type              | Drag coefficient C <sub>d</sub> |
|----------------------------|---------------------------------|
| General range              | 0.004 – 0.010                   |
| Axial-flow impeller        | 0.007                           |
| Radial-flow impeller (max) | 0.0095                          |

## 2.6 Kolmogorov (Microeddy) Length (λ)

Turbulence cascades energy down to ever-smaller eddies. Eddies of a size comparable to or smaller than a cell can damage it. Most mammalian cells are <20 μm.

$$\lambda = (v^3 / \epsilon)^{1/4} \text{ [m]}$$

v = kinematic viscosity (m<sup>2</sup>/s); ε = local energy dissipation rate (m<sup>2</sup>/s<sup>3</sup>). **Constraint:** keep λ > 20 μm for mammalian cells.

### 3. Gas-Sparging Parameters

Most cell death in bioreactors is caused by shear generated when bubbles **burst** at the liquid surface — not by the impeller. Sparging parameters therefore need their own constraints.

#### 3.1 Gas Volumetric Rate (VVM)

Volume of gas per unit volume of liquid per minute — the second most common scale-up criterion alongside P/V. Like P/V, it does not account for localized shear.

$$\text{VVM} = Q_{\text{gas}} / V_{\text{liquid}} \text{ [min}^{-1}\text{]}$$

#### 3.2 Gas-Entrance (Sparger-Hole) Velocity

Velocity of gas leaving each sparger orifice.

$$v_{\text{gas}} = Q_{\text{gas}} / (n_{\text{holes}} \cdot A_{\text{hole}}) \text{ [m/s]}$$

**Constraint:**  $v_{\text{gas}} < 30 \text{ m/s}$ ; above this, cell viability and productivity decline.

#### 3.3 Orifice Reynolds Number

Characterizes the bubble-formation regime at the sparger hole.

$$N_{\text{Re,orifice}} = (\rho_{\text{gas}} \cdot v_{\text{gas}} \cdot d_{\text{hole}}) / \mu_{\text{gas}}$$

Four regimes occur as flow increases: **discrete bubble** → **chain bubble** → **jetting** → **turbulent**. Near ~2,000 the system approaches the jetting regime, where bubbles are poorly dispersed by the impeller. **Constraint:** keep  $N_{\text{Re,orifice}} < 2,000$ .

#### 3.4 Bubble-Wake Eddy Length

EDR in the wake behind a rising bubble produces microeddies; as with impeller eddies, keep them  $> 20 \mu\text{m}$  to avoid harming cell growth.

#### 3.5 Bubble-Bursting Stress & Cell Death

Cell-death rate scales with the number of cells attached to the bubble-film volume per bioreactor volume. Surfactants (e.g., Pluronic) reduce cell–bubble attachment.

| Parameter                                     | Typical value           |
|-----------------------------------------------|-------------------------|
| Pluronic concentration                        | 1 g/L                   |
| Bubble film thickness ( $\delta$ )            | 50 $\mu\text{m}$        |
| Film-to-cell attachment factor ( $\epsilon$ ) | 0.4 per $10^6$ cells/mL |

**Sparger choice:** cells tolerate shear from drilled-hole or open-pipe spargers better than from microspargers (which create many small, high-curvature bubbles).

### 4. Design-Space Constraint Summary

The shear-proof operating window is the intersection of all of the following. A valid operating point must satisfy every row simultaneously.

| Parameter               | Method / basis      | Unit             | Constraint          |
|-------------------------|---------------------|------------------|---------------------|
| Average impeller EDR    | P/V                 | W/m <sup>3</sup> | ≤ 50                |
| Localized impeller EDR  | Impeller-zone calc. | W/m <sup>3</sup> | ≤ 1×10 <sup>5</sup> |
| Impeller Reynolds no.   | N <sub>Re</sub>     | –                | > 10,000            |
| Tip speed               | πND                 | m/s              | < 1.5               |
| Microeddy length        | Kolmogorov          | μm               | > 20                |
| Gas-entrance velocity   | Flow / orifice area | m/s              | < 30                |
| Orifice Reynolds no.    | Gas-velocity based  | –                | < 2,000             |
| Bubble-wake eddy length | Wake EDR            | μm               | > 20                |

## 5. Implications for the P/V Calculation Tool

The article points to a tool that does more than compute a single P/V number. Recommended capabilities:

- **Inputs:** impeller geometry (type, diameter D, power number N<sub>p</sub>), speed N, liquid density ρ and viscosity μ, working volume V, sparger geometry (hole count, hole diameter), and gas flow Q<sub>gas</sub>.
- **Core output:** P and P/V, with a pass/fail flag against the 50 W/m<sup>3</sup> guideline.
- **Validity check:** automatically compute N<sub>Re</sub> and warn the user if ≤ 10,000 (P/V scaling not valid).
- **Localized shear:** report impeller-zone EDR (using the ~70.5% power fraction for pitched-blade designs) against the 1×10<sup>5</sup> W/m<sup>3</sup> bound.
- **Derived metrics:** tip speed, Kolmogorov microeddy length, gas-entrance velocity, and orifice Reynolds number, each with its own pass/fail indicator.
- **Scale-up mode:** given a reference vessel, solve for the agitation speed at a new scale that holds P/V (or tip speed) constant, then re-check all constraints.
- **Design-space view:** visualize the feasible window where every constraint is simultaneously satisfied.

## 6. Conclusions from the Source

- P/V and VVM are the most widely used scale-up methods but ignore localized shear heterogeneity that is present at every scale.
- Multiple normalized engineering parameters are required to accommodate geometric and design differences among bioreactors.
- Operating conditions must simultaneously satisfy oxygen-transfer demand (cell respiration) and CO<sub>2</sub>-transfer demand (efficient pH control).
- A successful design space assesses both impeller and gas-sparging shear contributions — especially localized impeller shear for pitched-blade STBRs.

Disclaimer: This guidance summarizes a single trade publication for engineering reference. Equation coefficients and thresholds should be validated against primary literature and your own process data before use in GMP or design-of-experiment work.